

Duke Faculty Survey: Written Report

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Introduction

We used a probability sample to study faculty represented in 17 departments in Duke University's Trinity College of Arts & Sciences. We estimated the proportion who are women, the proportion with at least one grant ongoing in 2026, and, among Ph.D. holders, the average number of years since receiving the degree. Our fourth question estimated the proportion of Ph.D. holders who received their degree from Duke University.

Sampling Design

We constructed the frame from department-filtered Faculty results on Duke Scholars. It contained 1,152 unique faculty across 17 departments. Faculty with multiple affiliations appeared once and were assigned to the alphabetically first included department before outcomes were collected. Our target population is therefore faculty represented in this 17-department frame rather than all Trinity College faculty. Scholars served as a practical approximation to departmental rosters; incomplete or incorrect affiliations could cause coverage error.

We used proportionate stratified simple random sampling without replacement, with department as the stratum. We fixed the total sample at 230, approximately 20% of the frame, and allocated selections in proportion to department size using the largest-remainder method. We then sampled independently within each department using a fixed random seed.

Stratification ensured representation from all 17 departments and could improve precision because the proportion of women may vary by department. We did not use cluster sampling because reviewing individual profiles was the main collection cost, and we lacked an appropriate auxiliary size variable for probability-proportional-to-size sampling.

In department h , a faculty member's inclusion probability was $\pi_{hi} = n_h/N_h$, where N_h is the department frame size and n_h is its sample size. The survey weight was $w_{hi} = 1/\pi_{hi} = N_h/n_h$. Because the allocation was approximately proportional, weights were close to 5.

Data Collection

For each sampled faculty member, we collected information from Duke Scholars about whether the person was a woman, grants, and Ph.D. education. A faculty member was classified as having an ongoing grant when at least one listed grant, in any role, had dates that included 2026. No listed grant was coded as no ongoing grant.

We classified a person as a Ph.D. holder only when the education section explicitly listed a Ph.D. or D.Phil. We recorded the degree year and institution when available and calculated years since the degree

as 2026 – Ph.D. year. The fourth outcome indicated whether a Ph.D. holder received that degree from Duke.

To determine whether a faculty member was a woman, we first reviewed gendered pronouns on the person’s profile; this classified 95 faculty. When usable pronoun evidence was unavailable, we used the first name as an initial aid only if U.S. Social Security records for 1930–2005 associated it with one sex at least 95% of the time. We manually reviewed the remaining 28 cases, including uncertain or conflicting evidence and one pronoun match that referred to someone else. Every sampled faculty member received a final classification from the available profile information. Names, pronouns, and profile text are imperfect indicators, so some error remains possible.

Analysis

We used the `survey` package in R, specifying department as the stratum, inverse inclusion probabilities as weights, and department frame sizes as finite population corrections. Questions 1 and 2 used survey-weighted means of binary indicators. For Questions 3 and 4, we treated Ph.D. holders as a domain by subsetting the survey design. Question 4 used a logit confidence interval to keep its bounds between zero and one. All intervals have 95% confidence.

Results

1. **Faculty who are women.** We estimate that 41.5% of faculty represented in the frame are women (95% CI: 36.3% to 46.6%).
2. **Faculty with an ongoing grant in 2026.** We estimate that 31.9% have at least one ongoing grant in 2026 (95% CI: 27.2% to 36.7%).
3. **Years since receiving a Ph.D.** Among faculty identified as Ph.D. holders, the estimated mean is 27.8 years (95% CI: 25.8 to 29.8 years).
4. **Ph.D. holders who received the degree from Duke.** We estimate that 10.7% received their Ph.D. from Duke University (95% logit CI: 7.4% to 15.2%).

Limitations and Conclusion

Our estimates describe the 17-department frame and may not generalize to excluded or unlisted Trinity faculty. Scholars profiles may be incomplete or outdated, particularly for grants and education. Treating no listed grant as no ongoing grant may miss unlisted awards. Classification of whether a faculty member was a woman required judgment, and missing Ph.D. years or institutions could affect Questions 3 and 4. These sources of measurement and coverage error are not included in the confidence intervals.

Subject to these limitations, the sample suggests that 41.5% of faculty represented in the frame are women and 31.9% have an ongoing grant in 2026. Among identified Ph.D. holders, the estimated mean time since the degree is 27.8 years, and an estimated 10.7% received it from Duke.